SAMPLE COMPUTATION SHOWING MACHINE COMPARISON

Given: A series of ten (10) observations X_i and Y_i which have a known functional relationship of the form

$$y = ax + b$$

Then by the method of least squares we may solve for a and b.

$$a = \frac{n\Sigma X_i Y_i - \Sigma X_i \Sigma Y_i}{n\Sigma X_i^2 - (\Sigma X_i)^2} \quad , \quad b = \frac{\Sigma X_i^2 \Sigma Y_i - \Sigma X_i Y_i \Sigma X_i}{n\Sigma X_i^2 - (\Sigma X_i)^2}$$

The solution of this problem involves the following computations:

Computations	No.
Adds	40
Subtracts	2
Multiplies	26
Divides	2
Total	70

The following shows the computing time for the various computers:

7090	AN/FSQ-7A	STRETCH
5.1 msecs.	1.6 msecs.	0.2 msecs.

Stretch on least squares

i = 10

LX 0

L Xi
MPY Xi ~50 μs
ATM (ΣXi²)

(L Xi i
ATM (ΣXi)) ~25

(L Yi i
ATM (ΣYi)) ~25

(MPY Xi i
ATM (ΣXiYi)) ~42

ACBr

L (ΣXi)
MPY Xi
STO (ΣXiΣXi)

LD n
CMPY (ΣXi²)+(ΣXiXi)
STO (Denom) } 4.6

L (ΣXi)
MPY (ΣYi) } 3.5
STO (ΣXiΣYi)

LD n
CMPY-(ΣXiYi) } 4.0

D (Denom)) ~7.5
STO a

L (ΣXiYi)
MPY (ΣXi)

(LD (ΣXi²) ~4.0
CMPY-(ΣYi)

D (Denom)) ~7.5
STO b

29

{ advance V+1→V
count C-1→C
Branch

A
M
A
ACBr

7090 on heat square

CLA	2
FMP	13
FAD	7
STO	2
CLA X	2
FAD	7
STO IX	2
CLA Y	2
FAD	7
STO IY	2
LDQ	2
FMP	13
FAD	7
STO	2
TIX	2
	72 = (473. μs)×10 = 4730 μs

LDQ	2		
FMP	13		
STO	2		
LDQ	2	4730.	
FMP	13	382.	
FAD	7		
STO	2	5112. μs	7090
LDQ	2		
FMP	13	1632.	mid Cy
STO	2	~200 μs	stretch
LDQ	2		
FMP	13		
FSB	7		
FSH	18		
STQ	2		
LDQ	2		
FMP	13		
STO	2		
LDQ	2		
FMP	13		
FSB	7		
FSH	18		
STQ	2		
38 ops.	154. =	382. μs	